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Polypeptide having anti-aging activity, and application thereof

Abstract

Disclosed is a polypeptide capable of prolonging the lifespan of *Caenorhabditis elegans*, or pharmaceutically acceptable salts thereof. The polypeptide has an amino acid sequence XKFAA SEQ ID NO: 1) (X: any amino acid, preferably T/A). Further, after one or more amino acids on the foregoing polypeptide sequence are deleted, substituted, or added, the resulting polypeptide or the pharmaceutically acceptable salts thereof still have the anti-aging activity. The polypeptide has the effect of prolonging the lifespan of *Caenorhabditis elegans* and anti-aging, and also has the function of enhancing the ability of movement behavior, improving the anti-stress ability, and alleviating the decline of movement ability in the aging process.

Inventors:	Alphonse; Joseph Martinez Jean (Caux, FR), Xu; Hanmei (Nanjing, CN), Wang; Dong (Nanjing, CN)
Applicant:	NANJING ANJI BIOLOGICAL TECHNOLOGY CO., LTD. (Nanjing, CN)
Family ID:	72827293
Assignee:	NANJING ANJI BIOLOGICAL TECHNOLOGY CO., LTD. (Nanjing, CN)
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Primary Examiner: Steele; Amber D

Background/Summary

TECHNICAL FIELD

(1) The present invention relates to the field of biopharmaceuticals, and specifically, to a polypeptide having anti-aging activity, and application thereof.

BACKGROUND

(2) Aging is a process in which, after a period of sexual maturity, an organism weakens in self-renewal and repair capabilities of cells, degenerates in structure and function of tissues and organs, and eventually dies. It is characterized by diminished anti-stress ability, disturbance of balance, and increased risk of developing disease. With the aging of the global population, various degenerative diseases of the elderly and the immense medical expenses incurred have become more and more serious social problems. Maintaining the elderly population in healthy state is the key to reducing

the social and economic burden brought about by population aging. Therefore, it is of great significance to explore measures and methods for intervening in aging and aging-related diseases that are simple, economical, effective, safe, and suitable for popularization.

(3) In earlier research on aging, scientists believed that pure aging research should be strictly separated from aging-related disease research. Aging-related diseases generally refer to those of which the incidence increases with aging. At present, it is believed that aging-related diseases mainly include cardiovascular diseases, tumors, rheumatism, osteoporosis, cataracts, type 2 diabetes, hypertension, Alzheimer's disease, and the like. However, with the advancement in aging research, it has been found that the occurrence and development of various aging-related diseases are homologous to the aging process, and the aging process itself is the basic risk factor for many aging-related diseases. Social responsibility has also prompted biologists to realize that prolonging lifespan alone cannot reduce the heavy social and economic burden brought about by aging, and only prolonging healthy lifespan has practical significance. Therefore, how to reduce aging-related diseases, improve the quality of life of the elderly in old age, and prolong healthy lifespan has become a hot issue in aging research. At present, there is no strict indicator to measure healthy lifespan. It is generally believed that aging interventions, which can increase the body's anti-stress ability, reduce the occurrence and development of aging-related diseases, and alleviate aging-related degeneration, while prolonging lifespan, can be referred to as prolonging healthy lifespan.

(4) As a classic model animal for aging research, *Caenorhabditis elegans* has the following advantages. First, it has a relatively short lifespan, only 2-3 weeks under standard laboratory conditions, enabling analysis of whole lifespan survival; second, it is easy to obtain a large number of genetically identical animals under controlled ambient conditions; third, its transparent body allows us to directly observe how cells and tissues change with aging; and fourth, an insight into its cells and tissues, neuronal connections, and whole genomes contributes to anti-aging research. A major advantage of the *Caenorhabditis elegans* model is that biological information can be easily provided by genetic methods, thereby determining a large number of genetic mutations that can change lifespan. Finally, the lifespan of *Caenorhabditis elegans* exhibits remarkable plasticity and is affected by environmental conditions, nutritional conditions, and genetic mutations. The individual lifespan may change even under controlled conditions, thereby revealing random factors in aging. Therefore, *Caenorhabditis elegans* can be used as a model animal to study the anti-aging activity of drugs.

(5) By far, the prevailing anti-aging is the use of anti-aging drugs. Most of the anti-aging drugs used clinically are synthetic drugs. For example, vitamin E can promote cell division and inhibit the generation of oxygen free radicals, procaine preparations can prolong cell lifespan, and piracetam can delay brain aging; aspirin can delay the decline of body function caused by aging by combating oxidative stress, thereby prolong the lifespan of *Caenorhabditis elegans*; metformin, as an activator of AMP-activated protein kinase (AMPK), can also alleviate cognitive impairment and has a certain effect on delaying aging; and drugs such as PAL-12 (a hexapeptide) and resveratrol analogs are also favored for anti-aging.

(6) SUMMARY

(7) The Sequence Listing created on Mar. 31, 2022 with a file size of 2.00 KB, and filed herewith in ASCII text file format as the file entitled "Sequence_Listing-G204RAYT0007US.TXT," is hereby incorporated by reference in its entirety.

(8) 1. Problem to be Resolved

(9) The present invention provides a polypeptide with anti-aging activity, and application thereof. In the present invention, the polypeptide can effectively prolong the lifespan of *Caenorhabditis elegans*, which has a good anti-aging effect and has a great development prospect.

(10) 2. Technical Solutions

(11) To resolve the foregoing problem, the technical solutions adopted by the present invention are as follows:

- (12) A polypeptide or pharmaceutically acceptable salts thereof is provided, including an amino acid sequence of XKFAA (SEQ IN NO: 1), where X is H-Pro-Pro-Thr-Thr- (SEQ IN NO: 7), H-Pro-Thr-Thr-, H-Thr-Thr-, or any amino acid in a LID form, or acyl such as acetyl or propionyl.
- (13) For the polypeptide, X is T or A.
- (14) For the polypeptide, the amino acid sequence of the polypeptide is TKFAA (SEQ IN NO: 2) (named Ace) or AKFAA (SEQ IN NO: 3) (named RV1).
- (15) Use of the polypeptide in preparing anti-aging and/or life-extending drugs or supplements is provided.
- (16) The polypeptide can enhance the ability of movement behavior, improve the anti-stress ability, and alleviate the decline of movement ability in the aging process, thereby prolonging the average lifespan.
- (17) When necessary, one or more pharmaceutically acceptable adjuvants may also be added into the foregoing drugs, and the adjuvant includes a diluent, a filler, a binder, a wetting agent, an absorption enhancer, a surfactant, a lubricant, and a stabilizer that are conventional in the pharmaceutical field.
- (18) In the present invention, the drugs may be prepared into various forms of injection, freeze-dried powder injection, tablet, or granule. The drugs in various forms may be prepared by conventional methods in the pharmaceutical field.
- (19) 3. Beneficial Effects
- (20) Compared with the prior art, the present invention has the following beneficial effects: (1) The polypeptide of the present invention has a novel structure with the basic unit of natural amino acids, which is easy to synthesize, separate, and purify; (2) The polypeptide of the present invention can effectively prolong the lifespan of *Caenorhabditis elegans* and has the anti-aging activity; (3) The polypeptide of the present invention is safe with weak adverse reactions and toxic side effects, does not affect the growth and development of *Caenorhabditis elegans*, and does not affect the reproductive ability of *Caenorhabditis elegans*; and (4) The anti-aging effect of the polypeptide involved in the present invention has good performance in the *Caenorhabditis elegans* model, specifically embodied in significantly improving the ability of movement behavior of *Caenorhabditis elegans*, alleviating the decline of movement ability in the aging process of *Caenorhabditis elegans*, prolonging the time to half death of *Caenorhabditis elegans*, improving the anti-stress ability of *Caenorhabditis elegans*, and prolonging the lifespan of *Caenorhabditis elegans*.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 shows a survival curve of the experiment on the effect of polypeptide on the lifespan of *Caenorhabditis elegans*. It can be seen from the figure that the Ace and RV1 polypeptide groups have the effect of prolonging the lifespan of *Caenorhabditis elegans*.
- (2) FIG. 2A-2B show results of the effect of polypeptide on the body length and width of *Caenorhabditis elegans*. FIG. 2A shows the results of body length, and FIG. 2B shows the results of body width. The results are expressed as Mean $\pm$ SEM. Compared with the control group, n.s. represents no significant difference, * p <0.05 represents a significant difference, and ** P <0.01 represents a very significant difference. Compared with the control group, the polypeptide groups have no significant difference, indicating that the polypeptides Ace and RV1 do not affect the normal growth and development of *Caenorhabditis elegans*, and the polypeptides are safe.
- (3) FIG. 3 shows results of the effect of polypeptide on the number of offspring of *Caenorhabditis elegans*. The results are expressed as Mean $\pm$ SEM. Compared with the control group, n.s. represents no significant difference, * p <0.05 represents a significant difference, and ** P <0.01 represents a

very significant difference. Compared with the control group, the polypeptide groups have no significant difference, indicating that the polypeptides Ace and RV1 do not affect the reproductive ability of *Caenorhabditis elegans*, and the polypeptides are safe.

(4) FIG. 4A-4C show results of the effect of polypeptide on the ability of movement behavior of *Caenorhabditis elegans*. FIG. 4A shows the results of head swing, FIG. 4B shows the results of body bending, and FIG. 4C shows the results of frequency of pharyngeal pumping. The results are expressed as Mean±SEM. Compared with the control group, n.s. represents no significant difference, *P<0.05 represents a significant difference, and **P<0.01 represents a very significant difference. Compared with the control group, the polypeptide groups have a significant difference, indicating that the Ace and RV1 polypeptides can significantly enhance the muscle movement of *Caenorhabditis elegans* and improve the ability of movement behavior of nematodes.

(5) FIG. 5A-5B show a survival curve of the experiment on acute heat stress of *Caenorhabditis elegans*. FIG. 5A shows a survival curve of the experiment on acute heat stress of nematodes on Day 4, and FIG. 5B shows a survival curve of the experiment on acute heat stress of nematodes on Day 8. It can be seen from the figure that the Ace and RV1 polypeptide groups can significantly improve the ability of resistance to acute heat stress of nematodes, indicating that the polypeptide drugs can prolong the life cycle of nematodes in a thermal environment.

(6) FIG. 6A-6B show the results of rating of the movement ability of *Caenorhabditis elegans* in the experiment on acute heat stress. FIG. 6A shows the results of rating of the movement ability of nematodes in the experiment on acute heat stress on Day 4, and FIG. 6B shows the results of rating of the movement ability of nematodes in the experiment on acute heat stress on Day 8. It can be seen from the figure that, the proportion of nematodes in grade A in the Ace and RV1 polypeptide groups is higher than that in the control group, indicating that the polypeptide drugs can enhance the ability of resistance to acute heat stress of *Caenorhabditis elegans*, that is, the ability of movement behavior in a thermal environment.

(7) FIG. 7A-7B show the detection results of the experiment on acute oxidative stress of *Caenorhabditis elegans*. FIG. 7A shows the detection results of the experiment on acute oxidative stress on Day 4, and FIG. 7B shows the detection results of the experiment on acute oxidative stress on Day 8. The results are expressed as Mean±SEM. Compared with the control group, *P<0.05 represents a significant difference, and **P<0.01 represents a very significant difference. It can be seen from the figure that, the survival rate of nematodes in the Ace and RV1 polypeptide groups is significantly higher than that in the control group, indicating that the polypeptide drugs can prolong the life cycle of nematodes in an oxidative environment.

(8) FIG. 8A-8B show the results of rating of the movement ability of *Caenorhabditis elegans* in the experiment on acute oxidative stress. FIG. 8A shows the results of rating of the movement ability of nematodes in the experiment on acute oxidative stress on Day 4, and FIG. 8B shows the results of rating of the movement ability of nematodes in the experiment on acute oxidative stress on Day 8. It can be seen from the figure that, in the experiment on acute oxidative stress on Day 8, the proportion of nematodes in grade A in the Ace and RV1 polypeptide groups is higher than that in the control group, indicating that the polypeptide drugs can enhance the ability of resistance to acute oxidative stress of *Caenorhabditis elegans*; and in the experiment on acute oxidative stress on Day 4, there is no difference in the movement ability between the polypeptide groups and the control group.

(9) FIG. 9A-9C show the results of rating of the movement ability of *Caenorhabditis elegans* at different ages. FIG. 9A shows the results of rating on the movement ability of nematodes on Day 4, FIG. 9B shows the results of rating on the movement ability of nematodes on Day 8, and FIG. 9C shows the results of rating on the movement ability of nematodes on Day 12. It can be seen from the figure that, compared with the control group, in the Ace and RV1 polypeptide groups on Day 8 and Day 12, the proportion of nematodes in grade A is higher than that in the control group, indicating that the polypeptide can delay the muscle aging of *Caenorhabditis elegans*, and improve

the ability of movement behavior of nematodes, thereby prolonging the lifespan of nematodes.

(10) FIG. 10A-10F show the results of detection of the movement ability of *Caenorhabditis elegans* at different ages. FIG. 10A-B show the results of detection of the movement ability of nematodes on Day 4, FIG. 10C-D show the results of detection of the movement ability of nematodes on Day 8, and FIG. 10E-F show the results of detection of the movement ability of nematodes on Day 12. The results are expressed as Mean $\pm$ SEM. Compared with the control group, *P<0.05 represents a significant difference, and **P<0.01 represents a very significant difference. It can be seen from the figure that, compared with the control group, the frequency of head swing in the Ace and RV1 polypeptide groups is significantly higher than that in the control group, and the frequency of pharyngeal pumping on Day 12 is also significantly higher than that in the control group, indicating that the polypeptides can delay the muscle aging of *Caenorhabditis elegans*, improve the ability of movement behavior of nematodes, and alleviate the decline of movement ability of nematodes, thereby prolonging the lifespan of nematodes.

DETAILED DESCRIPTION

(11) The following further describes the present invention with reference to specific examples. The following descriptions are only preferred examples of the present invention, and are not intended to limit the present invention in other forms. Any person skilled in the art may use the technical contents disclosed below to change into equivalent examples with equivalent changes. Any simple modifications or equivalent changes made to the following examples according to the technical essence of the present invention without departing from the content of the solution of the present invention fall within the protection scope of the present invention.

(12) The polypeptide Ace (TKFAA, SEQ ID NO: 2) and polypeptide RV1 (AKFAA, SEQ ID NO: 3) were synthesized by the Engineering Research Center of Synthetic Polypeptide Drug Discovery and Evaluation of Jiangsu Province, with purities of 96.30% and 98.22%, respectively.

EXAMPLE 1

(13) Experiment on the Effect of Polypeptide on the Lifespan of the *Caenorhabditis elegans* model

(14) 1. Materials

(15) *E. coli* OP50 cultured at conditions: a shaking incubator, 220 rpm, and 37° C.

(16) Wild-type *Caenorhabditis elegans* cultured at conditions: a constant temperature and humidity incubator, 20° C., and humidity of 45-55%.

(17) 5-FUDR: 15.6 μ g/mL.

(18) NaN.sub.3: 0.4 M.

(19) H.sub.2O.sub.2: 30 mM.

(20) Ace: TKFAA (SEQ ID NO: 2), 10 nM.

(21) RV1: AKFAA (SEQ ID NO: 3), 10 nM.

(22) WD1: PPTTKFAA (SEQ IN NO: 4), 10 nM.

(23) WD2: PTTKFAA (SEQ IN NO: 5), 10 nM.

(24) WD3: TTKFAA (SEQ IN NO: 6), 10 nM.

(25) 2. Method

(26) Synchronization of *Caenorhabditis elegans*: The NGM plate with a moderate nematode density was selected. The larvae at the L4 stage were picked with a pick needle for about 20 in total and transferred into the blank NGM plate. After they developed into adults and laid the first batch of eggs, all the adults were picked out. About 12 hours later, the eggs hatched into larvae, to obtain the synchronized larvae at the L1 stage.

(27) Preliminary preparation for the experiment on the lifespan of *Caenorhabditis elegans*: The NGM was washed with 1 mL of M9 buffer. The resulting buffer containing the larvae at the L1 stage was collected into an EP tube, placed in a chromatographic cabinet at 4° C. for 5 min, and centrifuged at 1500 $\times$ g/4° C. for 3 min. The supernatant was removed. After 100 μ L of buffer was resuspended, the nematodes were counted under a microscope, and the nematodes were diluted to a concentration of 30 worms/10 μ L. 10 μ L of diluted nematodes were added into the NGM plate

to half death is 1 day longer than that in the control group. Based on the above, the Ace and RV1 polypeptide groups have better effects, so that the following anti-aging evaluation experiment is carried out on the Ace and RV1 polypeptides. Referring to Table 2 and FIG. 1 for details, the experimental results are statistically significant.

EXAMPLE 2

(35) Experiment of the Effect of Polypeptide on the Growth and Development of *Caenorhabditis elegans*

(36) 1. Materials

(37) The same as that of Example 1. 2. Method

(38) Synchronization of *Caenorhabditis elegans*: The same as that of Example 1.

(39) Experiment on the growth and development of *Caenorhabditis elegans*: The synchronized larvae at the L1 stage were collected and plated on the solid NGM inoculated with *E. coli* OP50. The nematodes were divided into a control group and a 10 nM dose group, and placed in a constant temperature and humidity incubator to culture for 72 h, and the body length and width of the nematodes in the adult stage were detected with a detection method as follows. 20 *Caenorhabditis elegans* were picked from each group and placed on a new solid NGM, and 50 μ L of 0.4 M NaN₃ was added dropwise. After most of the nematodes were rigid, the nematodes were photographed and recorded under an inverted microscope, and the body length and width of the *Caenorhabditis elegans* were measured with the ruler tool of Photoshop. One-way ANOVA statistical analysis was carried out on the data, and the data results were expressed as Mean $\pm$ SEM. Compared with the control group, *P<0.05 represents a significant difference, and **P<0.01 represents a very significant difference. 3. Experimental Results

(40) TABLE-US-00003 TABLE 3 Effect of polypeptide on the body length and width of *Caenorhabditis elegans*

Group	Control	Ace	RV1
Body length/ μ m	1256.974 $\pm$ 9.316	1255.680 $\pm$ 17.447	1249.000 $\pm$ 14.874
P value (body length)	1.000	0.997	
Body width/ μ m	63.647 $\pm$ 0.868	63.332 $\pm$ 0.964	61.763 $\pm$ 0.692
P value (body width)	0.999	0.549	

Note: The foregoing results are expressed as Mean $\pm$ SEM. Compared with the control group, *P < 0.05, and **P < 0.01.

(41) Compared with the control group, the body length and width of *Caenorhabditis elegans* in the polypeptide groups does not have a significant difference, which means that neither Ace nor RV1 polypeptides affect the normal growth and development of *Caenorhabditis elegans*, indicating that these polypeptides are safe. Referring to Table 3 and FIG. 2 for details, the experimental results are statistically significant.

EXAMPLE 3

(42) Experiment of the Effect of Polypeptide on the Reproductive Ability of *Caenorhabditis elegans*

(43) 1. Materials

(44) The same as that of Example 1. 2. Method

(45) Synchronization of *Caenorhabditis elegans*: The same as that of Example 1.

(46) Detection of the reproductive ability of *Caenorhabditis elegans*: The synchronized larvae at the L1 stage were collected and plated on the solid NGM inoculated with *E. coli* OP50. The nematodes were divided into a control group and a 10 nM dose group, and placed in a constant temperature and humidity incubator to culture to obtain the larvae at the L4 stage. 1 larva at the L4 stage was picked from each group and placed on a new solid NGM. It is ensured that the nematode was transferred to a new medium every day during the egg-laying period. The culture medium containing the eggs was cultured in an incubator for 24 h. The larvae on each plate were counted until the end of egg laying of the nematode. The data was summed up to obtain the total number of offspring of the nematode. The numbers of offspring of 10 nematodes were recorded for each group. One-way ANOVA statistical analysis was carried out on the data, and the data results were expressed as Mean $\pm$ SEM. Compared with the control group, *P<0.05 represents a significant difference, and **P<0.01 represents a very significant difference. 3. Experimental results

(47) TABLE-US-00004 TABLE 4 Effect of polypeptide on the number of offspring of *Caenorhabditis elegans* Group Control Ace RV1 Total number 289.70 ± 14.827 303.80 ± 13.012 278.50 ± 15.353 of offspring P value — 0.947 0.977 Note: The foregoing results are expressed as Mean $\pm$ SEM. Compared with the control group, *P < 0.05, and **P < 0.01.

(48) Compared with the control group, the total number of offspring of *Caenorhabditis elegans* in the polypeptide groups does not have a significant difference, which means that neither Ace nor RV1 polypeptides affect the reproductive ability of *Caenorhabditis elegans*, indicating that these polypeptides are safe. Referring to Table 4 and FIG. 3 for details, the experimental results are statistically significant.

EXAMPLE 4

(49) Experiment on the Effect of Polypeptide on the Ability of Movement Behavior of *Caenorhabditis elegans*

(50) 1. Materials

(51) The same as that of Example 1. 2. Method

(52) Synchronization of *Caenorhabditis elegans*: The same as that of Example 1.

(53) Detection of the ability of movement behavior of *Caenorhabditis elegans*: The synchronized larvae at the L1 stage were collected and plated on the solid NGM inoculated with *E. coli* OP50.

The nematodes were divided into a control group and a 10 nM dose group, and placed in a constant temperature and humidity incubator to culture for 48 h. Then, the following three indicators of the ability of movement behavior were detected: frequency of head swing, frequency of body bending, and frequency of pharyngeal pumping. For the frequency of head swing, 20 nematodes were picked from each group and transferred to a new and clean culture medium for acclimation for 1 h, a proper amount of M9 buffer was then added, and the number of times the nematode head swings from one side to the other and back within 30 s was observed and recorded under an inverted microscope. For the frequency of body bending, 20 nematodes were picked from each group and transferred to a new and clean culture medium for acclimation for 1 h, the number of times of body bending of the nematode within 30 s was observed and recorded under an inverted microscope, and the distance the nematode crawls forward by one wavelength was recorded as a body bending. For the frequency of pharyngeal pumping, 20 nematodes were picked from each group and transferred to a culture medium inoculated with *E. coli* OP50 for acclimation for 1 h. Under an inverted microscope, it was observed at sufficient magnification to clearly until see the pharyngeal pumping of the nematode, with photographing for 30 s for each nematode, and the number of pharyngeal pumping of the nematode was counted by slow playing video at speed of 0.3 times in PotPlayer. One-way ANOVAY statistical analysis was carried out on the data, and the data results were expressed as Mean $\pm$ SEM. Compared with the control group, *P<0.05 represents a significant difference, and **P<0.01 represents a very significant difference. 3. Experimental results

(54) TABLE-US-00005 TABLE 5 Effect of polypeptide on the ability of movement behavior of *Caenorhabditis elegans* Group Control Ace RV1 Frequency of head 56.60 ± 1.690 $67.65 \pm 1.708^{**}$ $62.60 \pm 1.162^{*}$ swing/30 s P value (head swing) — 0.000 0.040 Frequency of body 12.90 ± 0.397 $14.85 \pm 0.437^{**}$ $14.50 \pm 0.426^{*}$ bending/30 s P value (body bending) — 0.009 0.049 Frequency of pharyngeal 162.95 ± 2.318 $189.05 \pm 3.891^{**}$ $181.10 \pm 3.532^{**}$ pumping/30 s P value (frequency of — 0.000 0.000 pharyngeal pumping) Note: The foregoing results are expressed as Mean $\pm$ SEM. Compared with the control group, *P < 0.05, and **P < 0.01.

(55) Compared with the control group, the polypeptide groups have a significant difference in the ability of movement behavior of *Caenorhabditis elegans*, indicating that the Ace and RV1 polypeptides can significantly enhance the muscle movement of *Caenorhabditis elegans* and improve the ability of movement behavior of nematodes. Referring to Table 5 and FIG. 4 for details, the experimental results are statistically significant.

Claims

1. A polypeptide having anti-aging activity consisting of TKFAA (SEQ ID NO: 2), AKFAA (SEQ ID NO: 3), PPTTKFAA (SEQ ID NO: 4), PTTKFAA (SEQ ID NO: 5), or TTKFAA (SEQ ID NO: 6).
 2. The polypeptide according to claim 1, wherein each amino acid of the polypeptide is in a D or L form.
 3. A method for the preparation of anti-aging drugs or supplements, comprising providing a pharmaceutical composition comprising the polypeptide according to claim 1.
 4. The method according to claim 3, wherein one or more pharmaceutically acceptable adjuvants are also added into the pharmaceutical composition comprising the polypeptide, and the adjuvant comprises a diluent, a filler, a binder, a wetting agent, an absorption enhancer, a surfactant, a lubricant, and a stabilizer that are conventional in the pharmaceutical field.
 5. The method according to claim 4, wherein the pharmaceutical composition is in the form of an injection, a freeze-dried powder injection, a tablet, or a granule.
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